

Dimitris Tsitsigkos

Software Engineer (PhD)

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Work Experience

Postdoctoral researcher (Data Science and Engineering team)

Feb 2025 – Mar 2026

Archimedes Research on AI, Data Science and Algorithms, Athens, Greece

- Designed and implemented high-performance, in-memory multi-dimensional indices in C++, using parallelism and hardware-aware optimizations to improve query throughput and latency
- Technologies: C++, OpenMP, SIMD

PhD fellow

Jan 2024 – Dec 2024

Department of Computer Science & Engineering, Univ. of Ioannina, Greece

- Contributed to the implementation of spatial index structures in C++ for distance queries (k-NN, range, distance joins), improving query performance compared to existing approaches
- Contributed to the design and development of a prototype distributed spatial data management system using MPI and OpenMP, reducing memory consumption by over 55% while improving query performance.
- Technologies: C++, OpenMP, MPI

PhD fellow

Jul 2019 – Dec 2024

Information Management Systems Institute, Athena Research Center, Athens, Greece

- PhD Research:
 - Optimized B+-Tree Index Structures
 - * Implemented BS-tree, a gapped data-parallel B+-tree, and its compressed variant CBS-tree for high-performance in-memory indexing
 - * Both BS-tree and CBS-tree achieved $1.5\times$ – $2.5\times$ higher throughput versus B+-trees, learned indexes, and trie-based approaches across different read and update workloads
 - * CBS-tree uses 56%–94% less memory than competing indexes on compressible datasets while maintaining superior throughput
 - * Designed SIMD-friendly node layouts, branchless search, and efficient gap handling to fully exploit data-parallel CPU features, boosting single-threaded performance and scaling across cores
 - Indexing and Partitioning for Non-Point Spatial Data
 - * Developed two-layer space-oriented partitioning for indexing non-point spatial objects
 - * Reduced spatial join cost by $\sim 50\%$ compared to existing spatial indexing methods
 - * Achieved 2 – $3\times$ faster range query performance vs R-trees, quad-trees and grid-based indexes
 - * Designed techniques supporting parallel spatial query processing
 - Parallel Spatial Join Processing
 - * Implemented a parallel spatial join algorithm for rectangles using the Partition-Based Spatial Merge (PBSM)
 - * Optimized 1D/2D partitioning, sweep strategies, and data-driven parameter tuning to maximize join performance
 - * Achieved parallel spatial joins of tens of millions of rectangles in <1 s on multi-core systems, demonstrating high scalability and efficiency
- Participated in European funding project:
 - MORE: Management of Real-time Energy data.
 - * Built parallel continuous evaluation module in Java for sliding-window aggregations, enabling low-latency energy analytics
 - * Implemented parallel versions of time-series analytics algorithms in Python using Dask to improve efficiency and scalability for large-scale data processing
- Technologies: C++, OpenMP, SIMD, Python, Dask

Software Engineer

Apr 2017 – Nov 2018

Hellenic Army Information Technology Support Center, Athens, Greece

- Maintained and extended Java backend applications for internal systems, improving performance and implementing new features
- Technologies: Java, Oracle Database, JSF.

Research Engineer

Dec 2012 – Jun 2019

Information Management Systems Institute, Athena Research Center, Athens, Greece

Participated in several Greek and European funding projects, including:

- Amnesia: A platform for anonymizing relational, multi-dimensional, and hierarchical data.
 - Implemented an open-source data anonymization platform for in-memory anonymization, supporting both web and desktop applications
 - Designed the architecture and full stack, with a Java backend providing RESTful APIs (Spring) and a web frontend using JavaScript and HTML
 - Integrated the platform into the OpenAIRE ecosystem, enabling global researchers, data providers, and companies to anonymize data efficiently
- MoDisSENSE: A Distributed Spatio-Temporal and Textual Processing Platform for Social.
 - Built distributed spatio-temporal processing (Java/Hadoop/HBase) for trajectory reconstruction and POI analytics with RESTful APIs and PostgreSQL
- Technologies: Java, RESTful Web Services, Spring, MapReduce, Hadoop, HBase, PostgreSQL

Technical Skills

Languages:	C++, Java, Python
Data Management:	MySQL, PostgreSQL, PostGIS, HBase
Parallel & Distributed Systems:	OpenMP, SIMD, Spark, MPI, Hadoop, Dask
Backend & Web:	Spring, RESTful Web Services, JavaScript, JSP, JSF
Operating Systems:	Ubuntu, Microsoft Windows, macOS
Other:	Data management, in-memory indexing & optimization, parallel & distributed systems

Education

PhD, Data Management

Jul 2019 – Dec 2024

Department of Computer Science & Engineering, University of Ioannina (Uoi) in collaboration with Athena Research Center

- Visiting Researcher (PhD Collaboration), Institute of Computer Science (Data Management research group), Johannes Gutenberg University Mainz, Germany (May 2022 – Jul 2022)

Thesis: In-memory Indexing for Parallel Processing of Single and Multi-Dimensional Queries.

Published papers in: ICDE, VLDBJ, TKDE and SIG-SPATIAL

MSc, Computing Systems: Software and Hardware, Computer Science

Nov 2012 – Sep 2016

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

Thesis: Complex Event Processing (CEP) for Intrusion Detection.

BSc, Computer Science

Sep 2006 – Jun 2012

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

Thesis: Clustering Wikipedia resources.

Additional Experience & Awards

- [Best Research Paper Runner-Up](#) — ICDE 2026
- Published papers in top-tier peer-reviewed database conferences and journals (h-index: 5), including SIGMOD, VLDBJ, ICDE and EDBT
- 3rd place — [Future of Database Programming Contest](#), Athens (Mar 2025)
- Program Committee member and external reviewer for 8+ international conferences and journals in data management. Distinguished PC Member, ICDE 2026
- Volunteer: European Data Forum 2014, EDBT/ICDT 2023 Joint Conference, 6th ACM Europe Summer School on Data Science 2025, HDMS 2025

External Links

- LinkedIn: https://gr.linkedin.com/in/dimitris_tsitsigkos
- Google Scholar: https://scholar.google.com/citations?user=z_KVU0QAAAAJ
- DBLP: <https://dblp.org/pid/163/0537.html>
- GitHub: <https://github.com/dTsitsigkos>
- Personal website: <https://dtsitsigkos.github.io/>

Languages

- Greek (Native)
- English (Fluent)